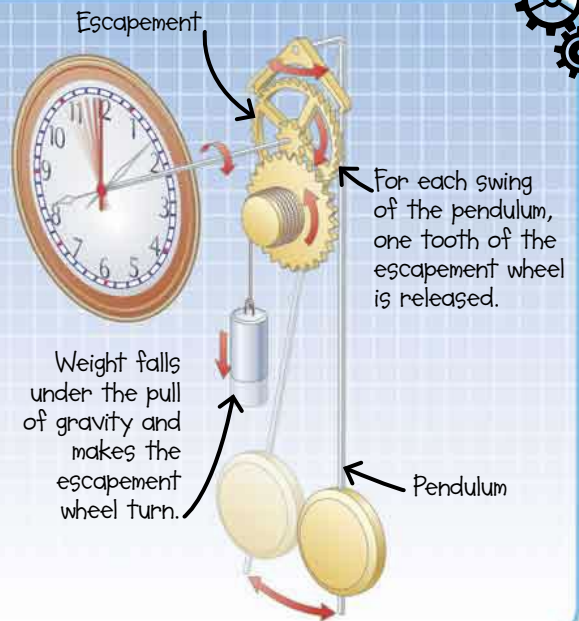


The time a pendulum takes to swing back and forth is always the same, as long as the length of the pendulum doesn't change and it keeps swinging. In a pendulum clock, the regular swing of the pendulum is captured by the escapement. The escapement is a device that uses the energy of the regular swing of the pendulum to allow the falling weight to move the hands on the clock face. At the same time, the escapement transfers energy from a falling weight to the pendulum to keep it swinging.



Marine chronometer

Pendulum clocks kept time on land, but at sea they were useless because of the repeated rocking of ships on the waves. Since **NAVIGATION** depended on telling time accurately, which could mean the difference between a successful voyage and a shipping disaster, governments offered a fortune in prize money to anyone who invented an accurate clock **without a pendulum**. In 1762, English carpenter **JOHN HARRISON** won the British government's prize with his *Number Four marine chronometer*.

Did you know?
Captain James Cook relied on Harrison's marine chronometer during his 1772 voyage from the Tropics to Antarctica.

Huygens's clock was the template for all pendulum clocks that followed. It used a falling weight to make the pendulum swing.



Marine chronometers used a balance wheel and a spring instead of a pendulum.

How it changed the world

Pendulum clocks remained the world's most accurate clocks for 300 years. Measuring time accurately not only meant that everyone could keep time, but also gave science an essential tool for experiments and research.

QUARTZ CRYSTALS, which vibrate at a constant rate in an electrical circuit, were first used in clocks in **1927**.



ATOMIC CLOCKS are the most precise timekeepers in the world. The first accurate one was made in **1955** by English physicist Louis Essen.

